

TASCO * VETC

Hackathon 2026

TASCO Atlas

The conversational map that
completes the journey

“Google Maps shows you where to go.
TASCO Maps completes the journey.”

01 Problem

02 Approach

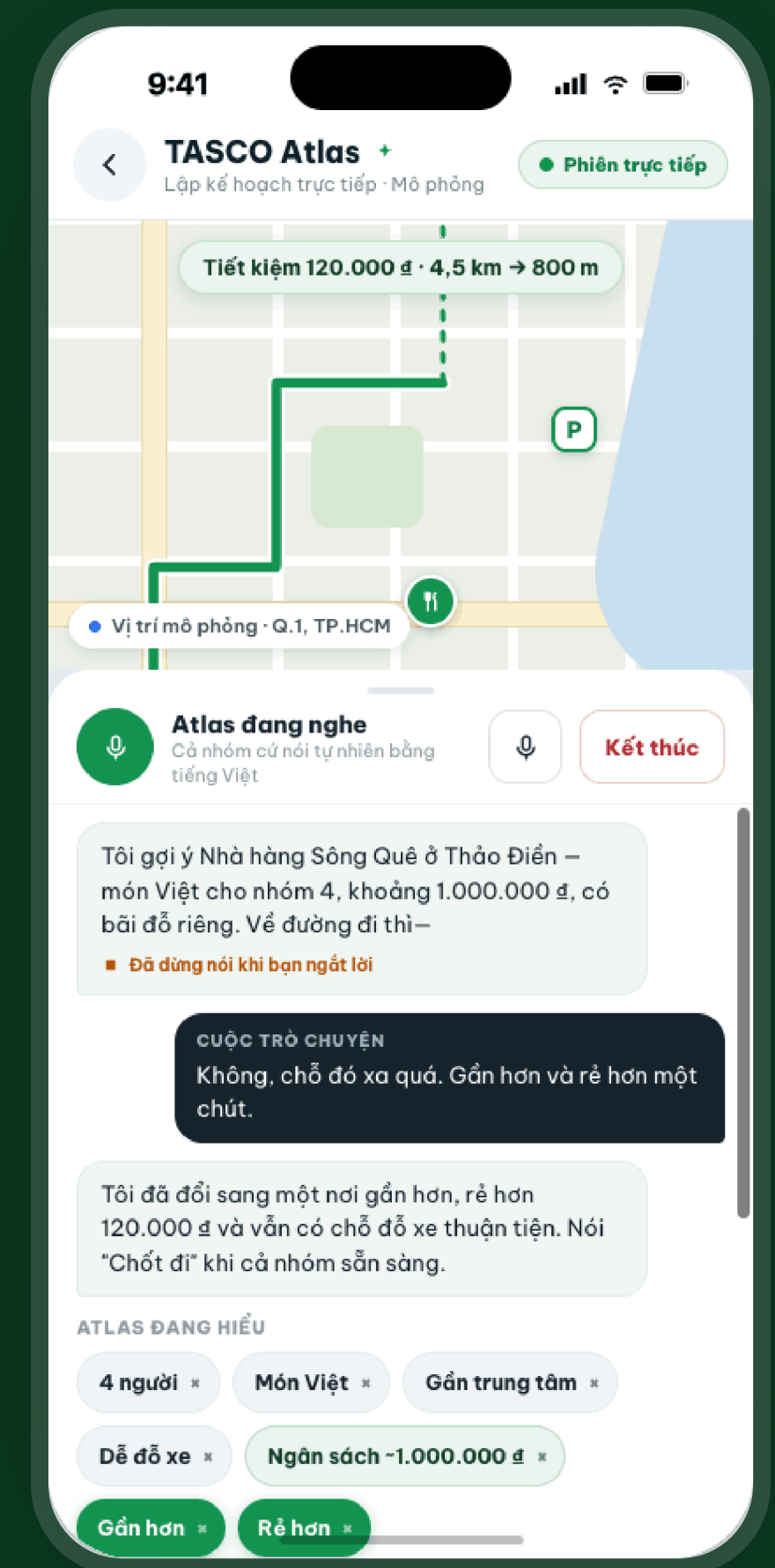
03 Live demo

04 Data & evaluation

05 Business model

06 5-week plan

Huy · Edward · Kiet



THE PROBLEM

Maps show places. Drivers still make every decision.

Today's maps are static search tools — every trip means searching, comparing, filtering, and rebuilding routes by hand.

“Dinner for three, around 1.000.000 ₺, somewhere easy to park.”

One sentence for a person — five searches and a dozen decisions for an app.

All of it lands on drivers, whose eyes and hands belong on the road — while VETC's own services sit fragmented across separate menus.

TODAY

Search



Compare places



Check parking



Find fuel



Estimate cost



Pay

WITH ATLAS

One natural conversation



One completed journey

OUR APPROACH

Talk naturally. Atlas turns intent into a journey.

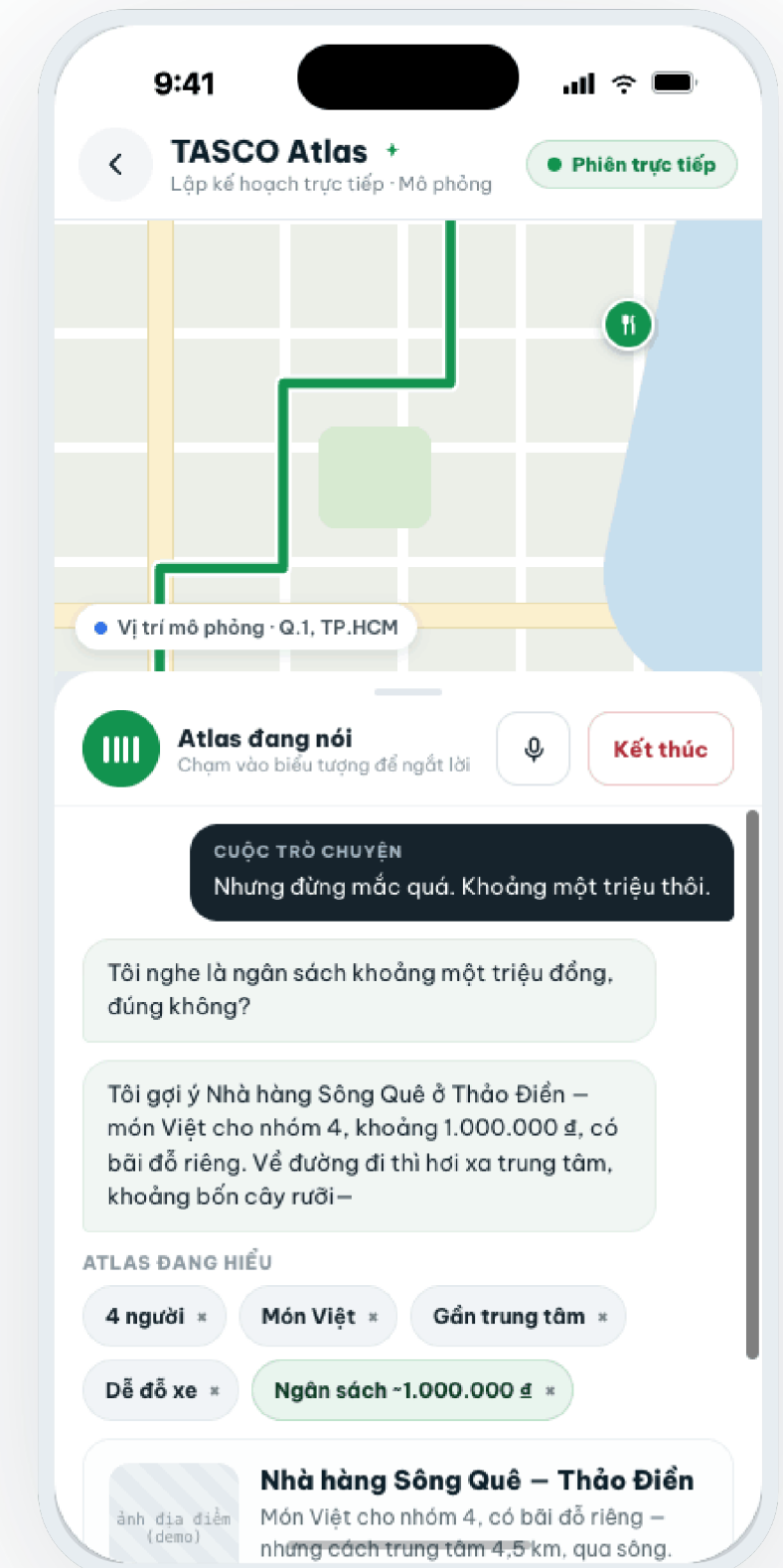
- 1 **Speak naturally** – Vietnamese, English, or mixed, hands-free.
- 2 **Atlas extracts constraints** – budget, party size, cuisine, parking, preferences.
- 3 **Searches and ranks the TASCO dataset** – and explains why each pick fits.
- 4 **Interrupt anytime** – *“Không pizza. Món Việt, gần hơn và rẻ hơn.”*
- 5 **Atlas recomposes instantly** – one journey, with the right VETC services attached.

Multi-turn memory

Grounded in the dataset

No camera

Session-only privacy



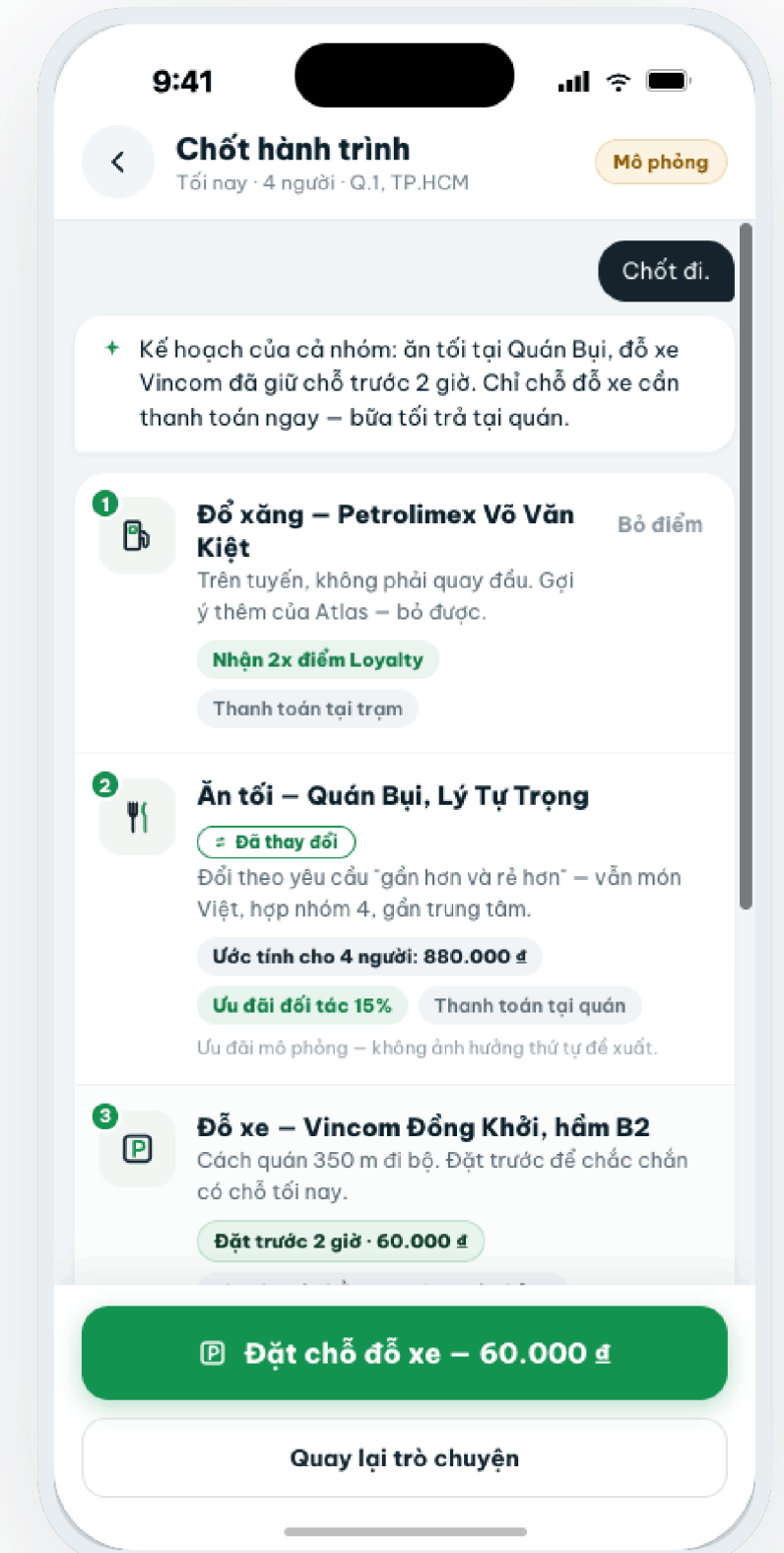
One request becomes an actionable route.

USER REQUEST

“Tối nay ba người muốn ăn món Việt, ngân sách khoảng 1 triệu, để đỗ xe.”

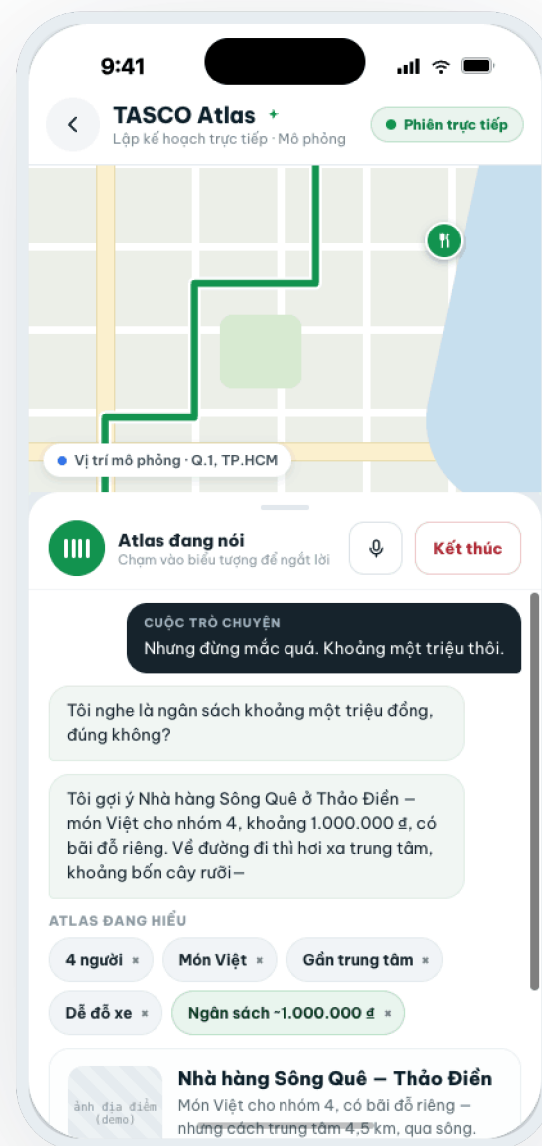
- 1 **Restaurant that fits** – with the reason it was chosen
- 2 **Fuel stop** – only when it's on the route
- 3 **Parking prebooked near the door** – the one prepaid item
- 4 **Route + VETC partner promotion** – reviewed on one screen

Mô phỏng – prototype prices, bookings and routes are simulated, and labeled as such

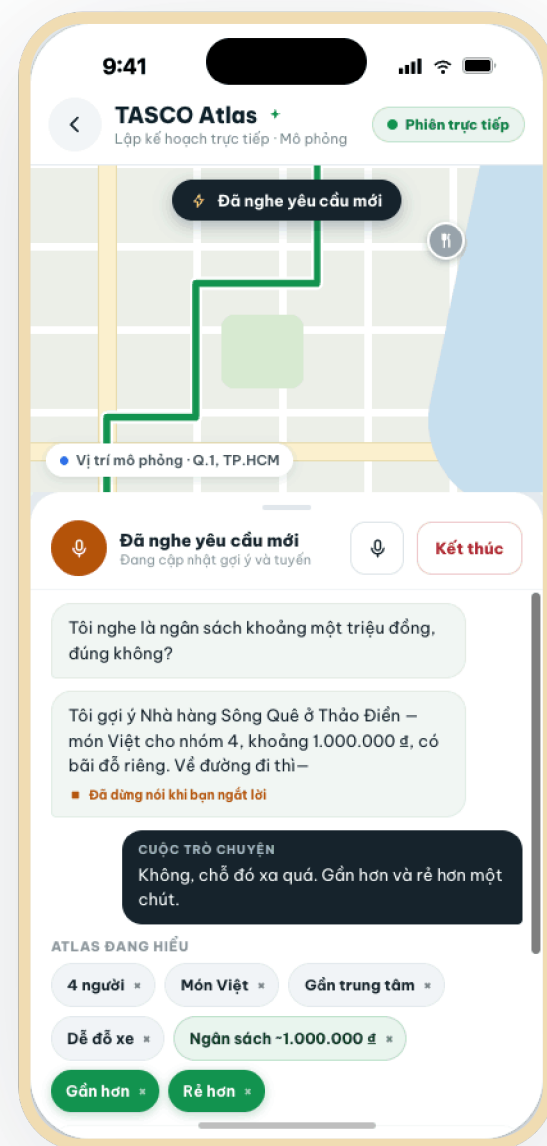


THE WOW MOMENT

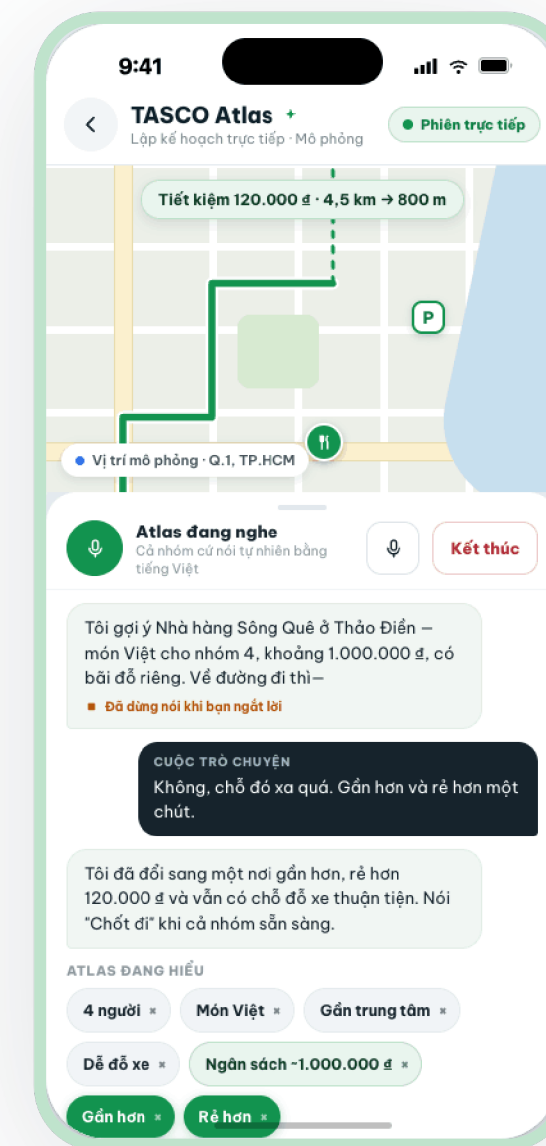
The map changes while the conversation is still happening.



1 · Atlas is mid-recommendation



2 · “Không, xa quá. Gần hơn và rẻ hơn một chút.”

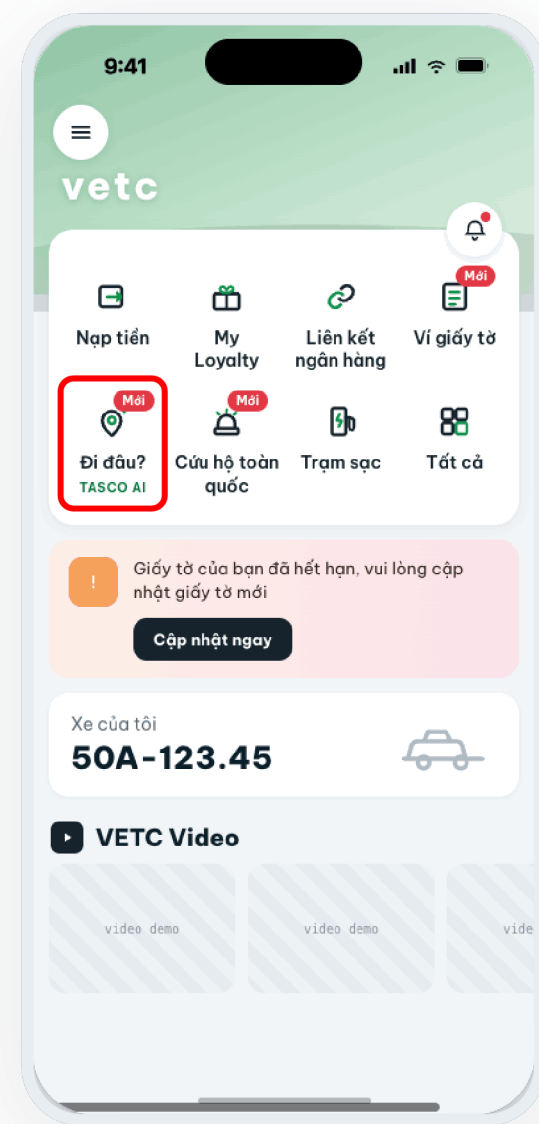


3 · Route redrawn — saves 120.000 đ · 4,5 km → 800 m

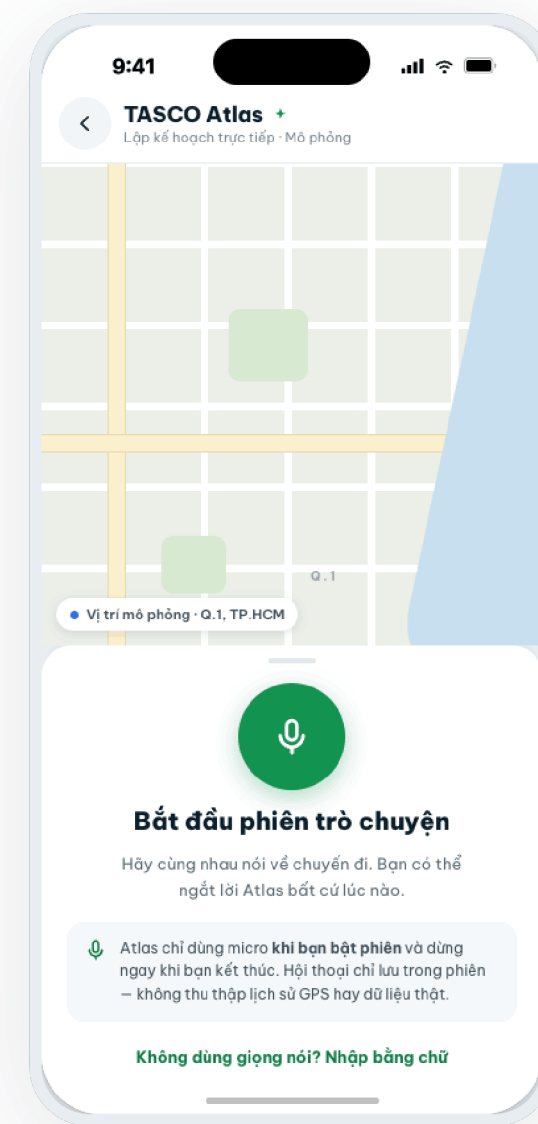
The wow isn't voice input — a shared conversation becomes a live, editable spatial plan, confirmed in the same-map 3D Route Theater.

WHAT WE BUILT

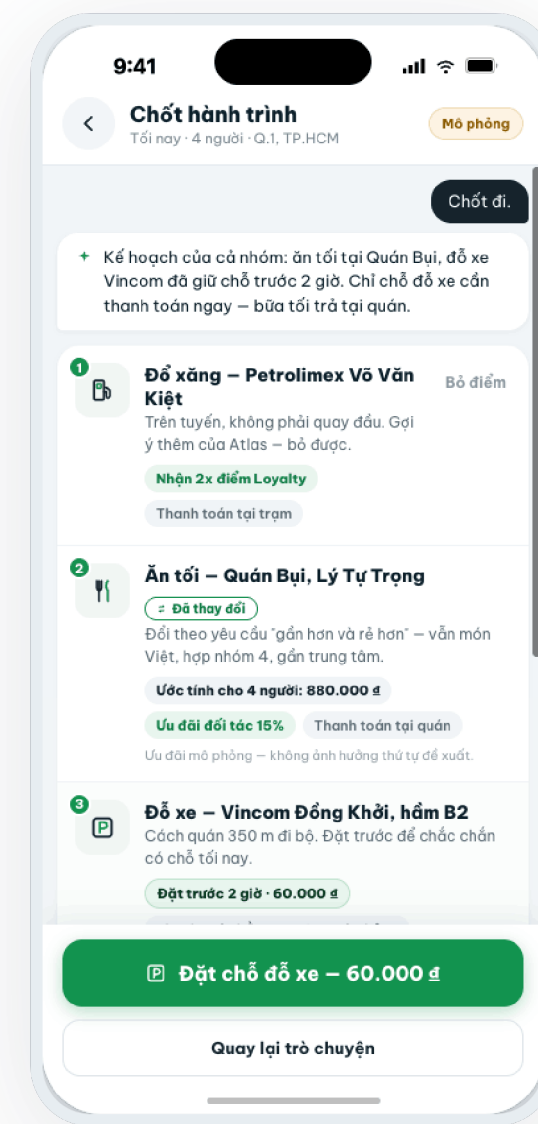
A working vertical prototype, not just a concept.



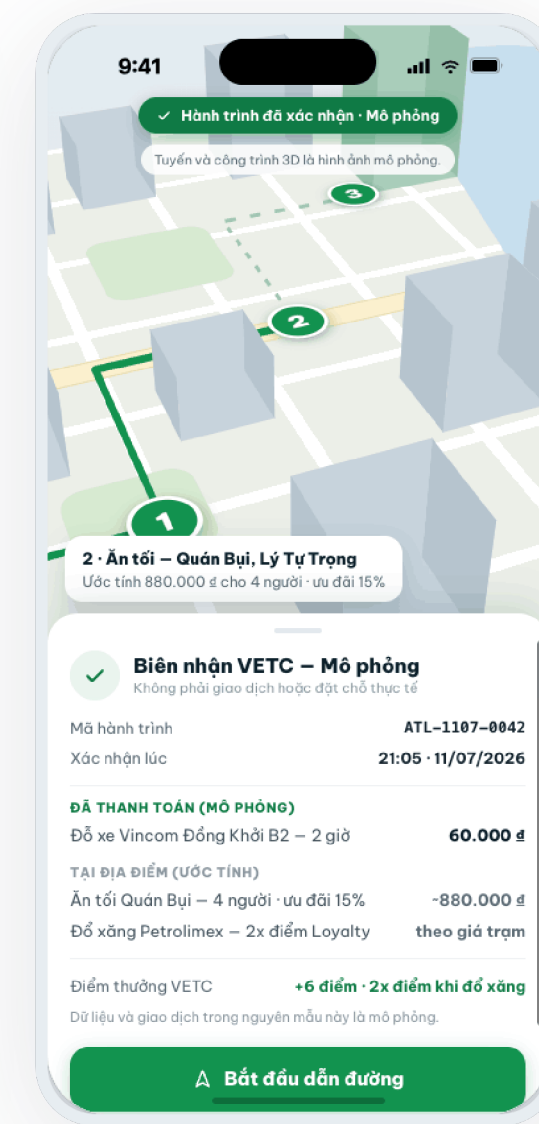
“Đi đâu?” on VETC home



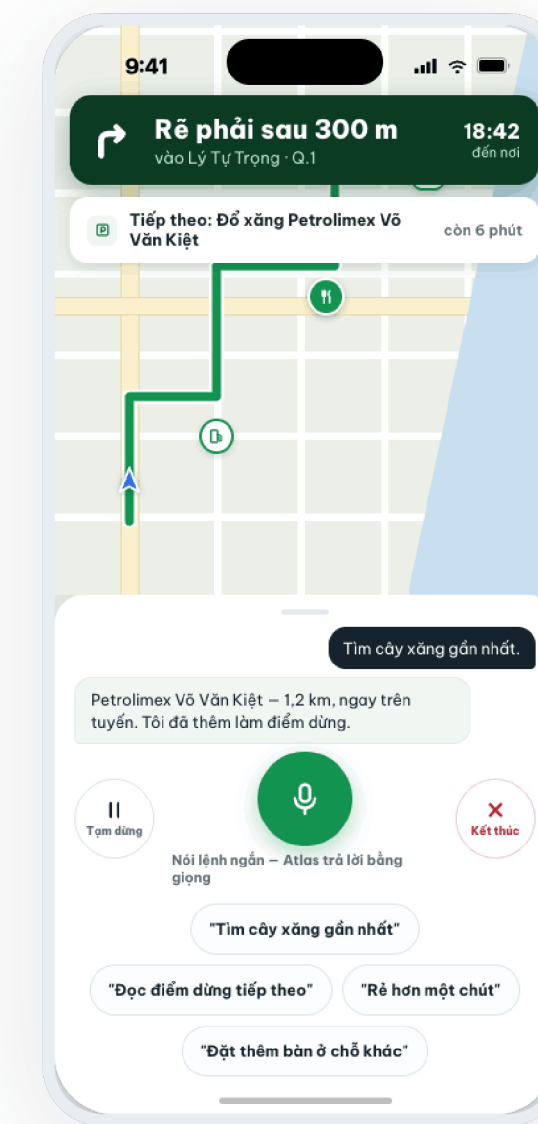
Voice start · explicit consent



Journey checkout



Receipt + 3D Route Theater



Safe driving mode

SYNERGIZES WITH VETC ECOSYSTEM

VETC Wallet · pays the journey

VETC Loyalty · points/vouchers

Mini App · hosts “Đi đâu?”

Insurance + Emergency · monitor the trip/call assistance

VETC Map · Scale with users

METHODOLOGY & EVALUATION

The official dataset was too weak to trust – so we built a real one.

OLD – OFFICIAL DATASET

8

samples – too weak to evaluate on

570×



NEW – OUR CONSTRUCTED DATASET

4,580

real-world samples, crawled for us by agents



TinyFish · browser-use agent



Overture · Maps Foundation

RESULT

>90%

accuracy on our constructed dataset

THE ENGINE

GPT Realtime handles the Vietnamese conversation – nothing else. The **deterministic journey engine** owns intent, ranking, prices and routes, so every recommendation is traceable.

PASSING TODAY

Ambiguous-place clarification · date-place recommendations · one-day itinerary planning · point-to-point routing

KNOWN, TRACEABLE GAPS

Category leakage · composite locations · numeric budget persistence · airport-as-location · one dataset gap (late-night food near Tân Sơn Nhất)

The official dataset had just 8 samples – ours has 4,580, and we score above 90% on it. Official scenario numbers land before the pitch. *(Eval in progress.)*

HOW TASCO MAKES MONEY

Every completed journey creates value inside the ecosystem.



1 · TRANSACTION REVENUE

Atlas recommends at the moment of intent. Users reserve or pay through VETC – TASCO earns a commission from restaurants, parking operators, charging stations and fuel partners, with promotions closing the loop in-app.

2 · ECOSYSTEM LIFETIME VALUE

Today users open VETC for one narrow task and leave. Atlas connects navigation to parking, fuel, charging, dining, insurance and loyalty – more useful journeys, more repeat usage, higher retention and LTV.

Future options: clearly-disclosed sponsored placements · merchant lead generation · loyalty-funded promotions · aggregated, consent-based demand insights.

SCALING THE EXISTING BASE

The users already exist – Atlas attaches spend to their trips.

TODAY

≈120,000

daily active users – team estimate

1–2 min

screen time – open, pay toll, close

0 đ

in-app spend beyond tolls



SCALED WITH ATLAS

Illustrative – BASECASE

5% attach one journey/day

6,000

journeys/day

avg bundle – fuel + meal + parking

250.000đ

blended take – commission + promos

8%

annual revenue to TASCO

43,8 tỷ đồng/năm

on 548 tỷ GMV flowing through VETC rails

Assumptions are adjustable live (Tweaks) · full funnel math and sensitivity in Appendix A11 · user-base figure to be verified before the pitch.

WHY TASCO CAN WIN

TASCO already owns the hardest starting point: the mobility relationship.

A driver-facing ecosystem with existing payment and service touchpoints – Atlas sits in the natural position between a driver's intent and the journey's completion.

Each completed journey makes the ecosystem more useful – a flywheel a standalone map cannot copy.

≈120,000 daily active users on VETC – team estimate, verify before the pitch.

ALREADY IN THE VETC ECOSYSTEM

Tolls (ETC)

Parking

EV charging

Roadside assistance

Insurance

Loyalty + payments

IF WE WIN – 5-WEEK EXECUTION PLAN

Week 1 is already shipped – four more weeks buys a live pilot.

WEEK 1 – ✓ DONE

Built this week

Live Vietnamese voice with barge-in · deterministic journey engine + checkout · driving mode · 2D/3D map · TinyFish + Overture pipeline → **4,580-sample dataset** + benchmark harness.

WEEK 2

Ship the pilot build

Close the remaining benchmark gaps + publish the eval report · VETC auth & consent · sandbox parking payment · edge-case hardening. **Artifact: pilot-ready build.**

WEEKS 3–4

Real data + driver beta

TinyFish agents refresh live hours & prices · real routing · new: journey history & loyalty earning · beta with 50–100 VETC drivers measuring attach rate, session length, voice completion. **Artifact: beta metrics report.**

WEEK 5

Deploy & scale

Staged rollout to a slice of VETC home traffic · live KPI dashboards · weekly cohort reviews · partner onboarding for dining & parking. **Artifact: live pilot + go/no-go memo for the TP.HCM launch – the first step toward 4.3M users.**

GROWTH PATH

120k DAU today → TP.HCM city launch → **4.3M VETC users nationwide** – all through the owned VETC funnel, zero paid acquisition.

MARKETING

“Đi đâu?” entry on VETC home + push at toll-payment moments; partner-funded promotions pay for users' first journeys.

BUDGET

\$9,500 total – voice API, data enrichment, infra, beta incentives. Full breakdown in **Appendix A12.**

Week 1 shipped the product – every following week ends with a measurable artifact: eval report, pilot build, beta metrics, live dashboards.

FROM SEARCHING FOR PLACES TO COMPLETING JOURNEYS

Less decision fatigue
for users

Safer, more natural
interaction for drivers

More transactions
and stronger retention
for TASCO

**“Google Maps shows you where to go.
TASCO Maps completes the journey.”**

Turn VETC from an app drivers open occasionally into the companion for every journey.

FOR Q&A

Appendix

A1 Product specification

A2 Architecture & technology

A3 API surface

A4 Recommendation methodology

A5 Context & history management

A6 Example conversations

A7 Journey Checkout mechanics

A8 User journey diagram

A9 Market size

A10 Unit economics & sensitivity

A11 Go-to-market & scaling to 4.3M

A12 Pilot budget — under \$10k

A13 Privacy & legal

A14 Risks & mitigations

A15 Judge FAQ

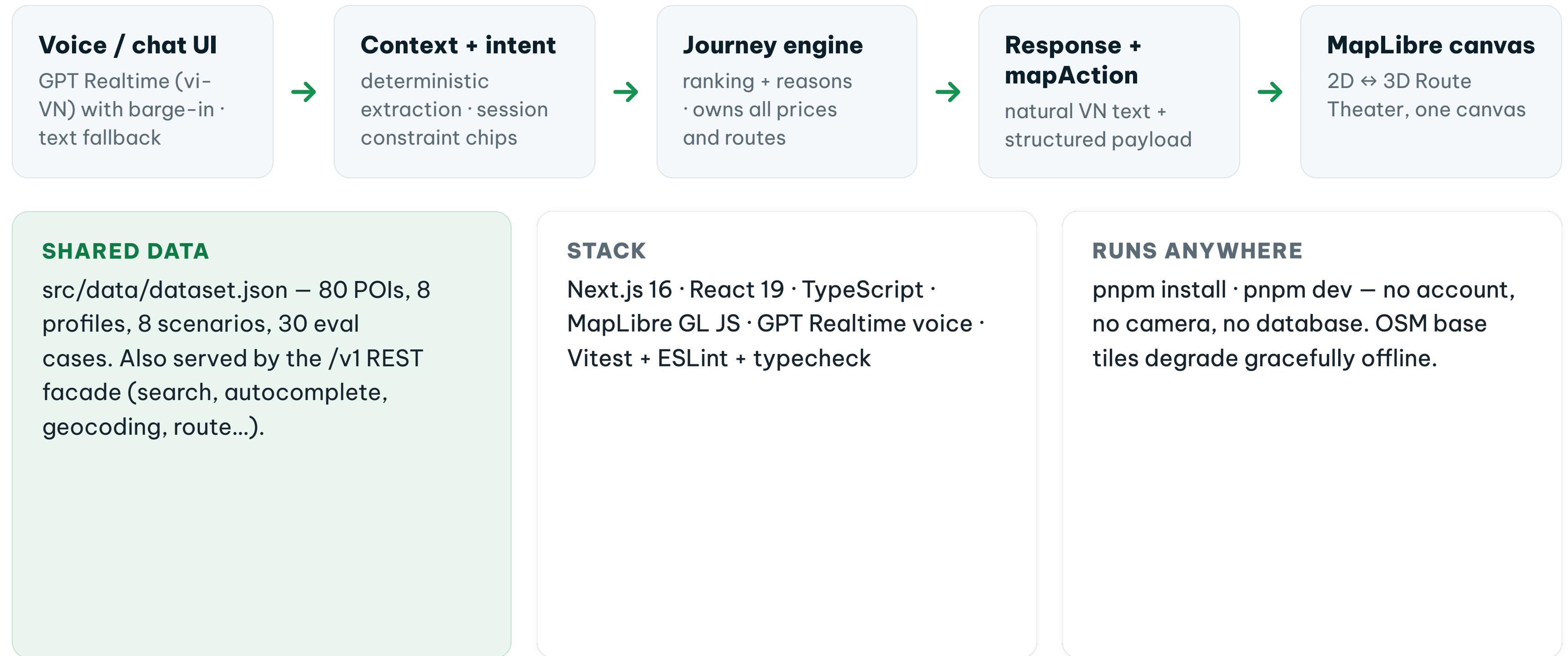
A16 Team & repository

Product specification vs. problem statement P8

REQUIREMENT (P8)	HOW ATLAS DELIVERS
Conversational search	Natural Vietnamese requests; normalization + alias handling; grounded in the TASCO dataset
Multi-turn interactions	Session context + constraint chips carried across turns – visible on screen (“Atlas đang hiểu”)
Clarification questions	Ambiguous entities return a clarify action with candidates – never a forced guess
Voice interaction	GPT Realtime live Vietnamese voice with barge-in interruption; explicit consent; text fallback
Personalized recommendations	Deterministic profile preferences / avoidances + budget compatibility (8 synthetic profiles)
Context understanding	Intent + entity extraction with session state; confidence attached to every response
Map action generation	Structured mapAction (search / navigate / clarify / journey) driving the MapLibre canvas live

Expected output format – intent, assistant response, recommendations with reasons, confidence – is returned verbatim by the API. Source: problem_statement.md, README.

Architecture – one application, no database



API surface – the supplied facade, fully implemented

METHOD	ENDPOINT	PURPOSE
GET	/health	Service health check
GET	/v1/search	Natural-language POI search and ranking
GET	/v1/autocomplete	Low-latency suggestions
GET	/v1/poi/{id}	Full POI details
GET	/v1/reverse-geocoding	Nearest POIs to coordinates
GET	/v1/nearby-search	Radius/category search around coordinates
GET	/v1/geocoding	Resolve place/address to dataset coordinates
POST	/v1/route	Deterministic simulated route + alternatives
POST	/api/chat	Conversation, recommendation, context, map action

Full OpenAPI 3.1 contract at `public/openapi.yaml`, served live at `localhost:3000/openapi.yaml`.

Recommendation methodology – deterministic and explainable

Blended ranking signals

Vietnamese normalization + aliases – diacritics, brands, place nicknames

Category & location inference – from the query and conversation context

Profile preferences / avoidances + budget – per synthetic user profile

Haversine proximity – when coordinates are provided

Rating & popularity blend – with stable tie-breaking

EVERY RESULT SHIPS WITH

a human-readable reason · a confidence score · a weighted score breakdown

ROLE OF THE LLM

GPT Realtime carries the Vietnamese conversation and barge-in. The deterministic pipeline stays authoritative for intent, ranking, map actions, prices and totals – and is the automatic fallback when no model is available.

Context & conversation history – session-only, visible on screen

Carried across turns

Each response returns a `sessionContext`; the client passes it – plus the history array – into the next request. No server session store.

Constraint chips users can see – and remove

"Atlas đang hiểu" shows the live constraints (4 người · Món Việt · ~1.000.000 đ · dễ đỗ xe). Removing a chip removes the constraint – memory is inspectable, not hidden.

Ambiguity → clarify, never guess

Ambiguous entities produce a `clarify` action with explicit candidates; the chosen candidate updates the session context.

Privacy boundary

Every response is labeled `privacy: session-only, persisted: false`. No durable conversation database exists anywhere in the system.

Example conversations – three of the ten-plus documented

intent: recommendation

"Tối nay ba người muốn ăn món Việt, ngân sách khoảng 1 triệu, dễ đỗ xe."

A ranked plan – restaurant with reason, on-route fuel, prebooked parking – reviewed on one screen.

intent: barge-in revision

"Không, chỗ đó xa quá. Gần hơn và rẻ hơn một chút."

Atlas stops mid-sentence, keeps every other constraint, swaps one stop, redraws the route – saving 120.000 đ.

intent: clarify

"Đưa tôi đến Galaxy."

Atlas asks which Galaxy – Cinema, Hotel, or Shopping Center – rather than inventing a destination.

Journey Checkout mechanics – grounded, priced, honest

Eligibility & composition

A driving request naming at least two of **fuel, dining, parking** produces one journey with 2–3 dataset-grounded actions and stable action IDs.

Exact VND arithmetic

Integer VND prices, discounts, loyalty rewards and totals computed locally – sponsorship disclosed in-line, simulated: true on every action.

Honest revisions

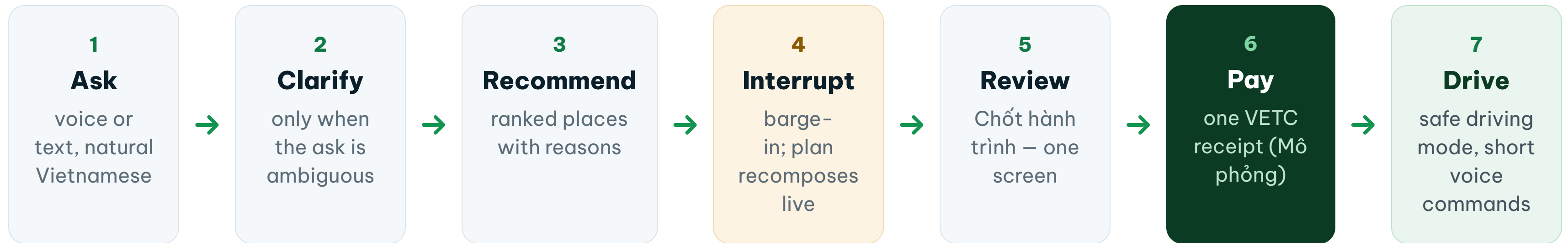
"Rẻ hơn một chút" either returns a strictly cheaper same-category bundle – or explicitly keeps the current journey. No fake savings, ever.

One receipt, then the show

Confirmation issues a single idempotent, simulated VETC receipt (Biên nhận VETC – Mô phỏng), then hands off to the 3D Route Theater to play the journey.

Only prepaid item in the demo journey: parking prebooking (60.000 đ) – dinner is paid at the venue, fuel at the station.

User journey – one question to a completed, paid trip



Every step happens inside one conversation on one map canvas. Discovery, decision, and payment never leave the app – that is the behavioral difference from a directions-only map.

Market size — payment is already on file for millions of drivers

TAM

Roadside spend of Vietnamese car trips

Fuel + food & drink + parking attached to private-car journeys nationwide — tens of nghìn tỷ đ per year, directional.

SAM

VETC-tagged vehicles

Millions of vehicles with an e-tag and a funded wallet — identity, vehicle and payment already on file. No checkout friction to solve.

SOM

≈120,000 daily active users today

The beachhead the money math is built on. Growth comes from the existing VETC funnel, not paid acquisition.

Directional public estimates — not audited; verify figures before external use.

Unit economics – the 43,8 tỷ đ math, step by step

120.000 DAU × 5% attach



6.000 journeys/day



× 250.000 đ bundle × 365



547,5 tỷ GMV/năm



× 8% take ≈ 43,8 tỷ đ/năm

CONSERVATIVE

2% attach · 250k bundle · 5% take

≈ 11 tỷ đ

per year – still meaningful on owned rails

BASE CASE

5% attach · 250k bundle · 8% take

≈ 43,8 tỷ đ

per year – the number on slide 10

UPSIDE

10% attach · 300k bundle · 10% take

≈ 131 tỷ đ

per year – before growing past 120k DAU

Illustrative – Mô phỏng. Assumptions adjustable live in Tweaks; DAU figure to be verified with TASCO.

Go-to-market – an owned funnel to 4.3M VETC users

PHASE 1 · PILOT – WEEKS 3–5

50–100 drivers

Hand-recruited VETC drivers in TP.HCM. Prove the gate metric: **≥5% of trips attach a journey**. Weekly cohort reviews; kill or scale on data.

PHASE 2 · CITY – MONTHS 2–3

TP.HCM launch, ~120k DAU

Feature “Đi đâu?” on VETC home for the existing daily base. Partner-funded promotions (free parking hour, fuel discount) reward the first completed journey.

PHASE 3 · NATIONAL – MONTHS 4–12

All 4.3M VETC users

Push at toll-payment moments – the one time every driver opens the app. Expand partner supply city by city; loyalty points close the repeat-usage loop.

CAC ≈ 0 – every target user already has the app, an identity, a vehicle and payment on file. Marketing spend goes to partner-funded first journeys, not ads.

Pilot budget – \$9,500 all-in, under the \$10k cap

LINE ITEM	WHAT IT BUYS	EST. USD
GPT Realtime voice API	5 weeks of dev + beta voice sessions, Vietnamese-first	\$3,000
Data enrichment	TinyFish agentic extraction runs – hours, menus, prices (Overture is free & open)	\$1,500
Hosting & infra	Staging + pilot deployment, monitoring dashboards	\$800
Driver beta incentives	100 drivers × ~\$25 in journey credit (partner co-funded where possible)	\$2,500
Launch assets	In-app creative for the “Đi đâu?” entry + push templates	\$700
Contingency	~12% buffer for API overage and re-runs	\$1,000
TOTAL	Team labor contributed in-kind by Huy, Edward, Kiet	\$9,500

Estimates, not quotes – refresh vendor pricing before committing. VETC integration engineering assumed to come from the TASCO side.

Privacy & legal – consent-first by design

Consent-first voice

The microphone starts only on an explicit tap with visible consent copy. No always-on listening, no wake word, no camera – and a full text fallback.

Session-only memory

Every response is labeled privacy: `session-only`,
`persisted: false`. No durable conversation database exists
– constraints die with the session.

Vietnam PDPD alignment

Designed to align with **Decree 13/2023/NĐ-CP** – explicit consent, purpose limitation, no sale of personal data. Formal legal review scheduled in the integration week.

Honest commerce

Sponsored placements always disclosed in-line; every simulated price, booking and receipt carries the *Mô phỏng* label until real rails are wired.

Risks & mitigations – what could go wrong, and the plan

Attach rate lower than 5%

Mitigation: conservative case already planned at 2% (still ≈11 tỷ đ/năm). Week-5 pilot gate decides scale-up on measured data, not hope.

Voice accuracy in a noisy car

Mitigation: safe driving mode with short canned commands, confirmation before any spend, and a full text fallback when confidence drops.

Partner supply – restaurants, parking

Mitigation: start with VETC's existing parking, fuel and charging partners – already contracted. Dining joins via lead-gen promotions, not upfront integration.

LLM cost & latency at scale

Mitigation: the deterministic engine does the heavy lifting – the model only converses. Cost is capped in the pilot budget and measured per journey before scaling.

Judge FAQ – the six questions we expect, answered

“Why won’t Google just build this?”

They don't own the transaction rails, the e-tags, or the partner contracts – TASCO does. The moat is the ecosystem flywheel, not the map. → [Slide 11](#)

“Are the prices real?”

The deterministic engine owns every price – today simulated and labeled Mô phỏng; weeks 3–4 wire sandbox payments and TinyFish live prices. → [A7](#)

“What’s real in the demo?”

Live Vietnamese voice with barge-in, deterministic ranking, the official dataset, 41 passing tests. Simulated: routes, bookings, payments. → [A2](#)

“Is voice while driving safe?”

Explicit-consent start, a driving mode with short commands, confirmation before spend, and no screen interaction required mid-route. → [A13](#)

“What if only 2% attach?”

Still ≈ 11 tỷ đ/năm on rails TASCO already operates – and the week-5 gate stops spend before scaling if the data disagrees. → [A10](#)

“Why 4/8 and not 8/8?”

Because we measured honestly. Every gap is known and traceable, and week 1 targets 8/8 with the test suite grown past 100. → [Slide 8](#)

Team & repository

Huy

Problem framing, product story, pitch
– opens the deck and owns the vision.

Edward

Engineering – voice layer, journey
engine, evaluation benchmark,
live demo.

Kiet

Business model, unit economics, go-
to-market and the execution plan.

EVERYTHING IS INSPECTABLE

One repository: `pnpm install · pnpm dev` – no account, no database. Docs for architecture, methodology and example conversations · OpenAPI 3.1 contract at `public/openapi.yaml` · the published eval dataset with full traces for all 8 official scenarios.